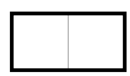
Building Bigger Shapes from Smaller Ones

counting the small squares in a composed figure, naming the new shape, comparing two builds, and finishing a rectangle

Directions

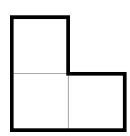
- Small shapes can be fitted together to build a bigger shape. The pieces must not overlap and must leave no gaps.
- The thin lines show the small squares. The thick line shows the outside of the built shape.
- Touch each small square once as you count, so none is counted twice.

1. How many small squares cover this shape?



Number of small squares: _________

2. How many small squares cover this shape?

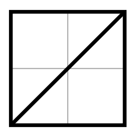


Number of small squares: _________

3. How many small squares cover this rectangle?

Number of small squares: _________

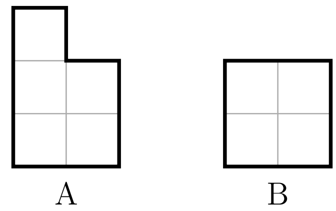
4. Two matching triangles are joined along their longest edges. The thick line shows where they meet.



(a) Name the new shape the two triangles make. _________

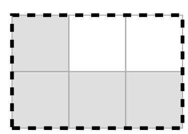
(b) How many small squares does the new shape cover? _________

5. Two shapes were built from small squares.



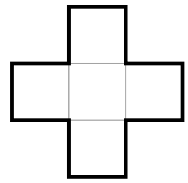
Which shape uses more small squares? Write <, >, or =: A ________ B

6. The dashed line shows the rectangle to be built. The shaded squares are already in place.



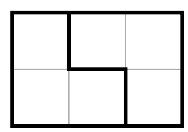
How many more small squares are needed to fill the rectangle? ________

7. How many small squares cover this shape?



Number of small squares: ________

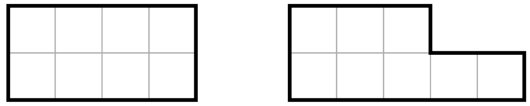
8. Two matching corner pieces are fitted together. The thick line inside shows where they meet.



(a) Name the new shape the two pieces make. ________

(b) How many small squares does the new shape cover? ________

9. Two shapes were built from small squares. They do not look alike.



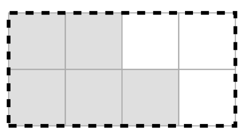
A B

(a) Count the small squares in each shape. Write <, >, or =: A _________ B

(b) Explain how two shapes that look so different can use the same number of small squares.

---

10. The dashed line shows the rectangle to be built. The shaded squares are already in place.



(a) How many more small squares are needed to fill the rectangle? _________

(b) Explain how you worked that out without placing the squares one at a time.
